Case Study: Bridging the Gap

How Accessibility HMI Reduces Collision Risk through Multimodal Feedback
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GitHub Repository: https://github.com/nina344/hmi-accessibility-prototypes

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1. Executive Summary

This research demonstrates that **Accessibility (a11y)** in automotive design is not just about inclusivity for users with permanent disabilities—it is a **critical safety layer** for every driver. By implementing **multimodal feedback (haptic and ambient)**, we can bypass **cognitive load** and visual distraction, effectively reducing **perception-reaction time**. Through **kinematic modeling** and exponential risk analysis, this study indicates that a **0.5-second improvement** in reaction time can reduce the **collision risk factor by ~4.5x.**

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2. Problem Statement: The Distracted Driver

In real-world scenarios (night driving, fatigue or cognitive distraction), a driver's visual channel is often compromised. At a standard speed of **60 km/h (~16.67 m/s)**, a distracted driver might take up to **2.0 seconds** to recognize a hazard and initiate braking.

- **Initial Distance to Hazard (Deer):** 20 meters.
- **Hazard velocity (Deer):** ~5 m/s (crossing trajectory), resulting in a **relative closing speed of 11.67 m/s**.
- **Result:** The vehicle travels a **relative distance of 23.34 meters** (closing the gap with the hazard) before the driver even touches the brake.
- **Outcome:** A high-velocity **collision is mathematically inevitable**.

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3. The Solution: Multimodal HMI Prototype (PoC)

To address visual distraction, I developed a **Proof of Concept (PoC)** that transitions the HMI from a **SAFE state to a DANGER state** using **multimodal, non-visual cues**.

This architecture ensures that critical safety information reaches the driver even when their visual focus is elsewhere.

System Architecture & Alert Flow:

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Hazard Detected → HMI State: SAFE → DANGER
                  ├── Haptic: Directional seat vibration
                  ├── Ambient: Red pulse (peripheral awareness)
                  └── Override: Secondary info suppression (priority logic)
```

Key Interaction Channels:

- **Directional Haptic Feedback:** Vibrations in the seat provide **immediate spatial awareness** without requiring the driver to look at the dashboard.
- **Ambient Peripheral Lighting:** Pulse effects trigger **subconscious responses** faster than traditional icons, utilizing the driver's peripheral vision.
- **Logic Arbitration:** As demonstrated in my GitHub repository, the system prioritizes **safety-critical alerts**, ensuring the driver is "woken up" instantly by **suppressing non-essential UI elements.**

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4. Mathematical Proof & Kinematics

By making the hazard "accessible" through touch and peripheral light, we reduce the **reaction time to 1.0 second**.
With a **relative closing speed of 11.67 m/s** (as established in Section 2) and an **emergency braking rate of -6.0 m/s²**, after the 1.0s reaction phase the vehicle has **8.33m** of the initial gap remaining.

Kinematic Equation for Collision Point:
We use the quadratic formula to solve for ***t*** **(Time to Collision**) after braking begins, derived from the displacement formula **($d = v \cdot t + 0.5 \cdot a \cdot t^2$):**

$$3t^2 - 11.67t + 8.33 = 0$$

- 3 represents 0.5 · a in **m/s²**
- 11.67 is the relative velocity in **m/s**
- 8.33 is the remaining distance gap in meters **(m)**

Results:

- **Standard Reaction (2.0s):** Impact occurs **before braking, at ~1.7s**.
- **A11y HMI Reaction (1.0s):** Collision is **delayed to 1.94s**, with the vehicle significantly decelerated.

Risk Reduction Model:

We model the probability of a collision using the **exponential risk function:**

$$\mathbf{risk = e^{k \cdot (threshold\text{-}TTC_{effective})}}$$

To ensure reproducibility, the following constants were applied:

- **Sensitivity Constant (k): 3** - Models the steep, non-linear escalation of risk as reaction time decreases, reflecting real-world conditions where small delays near the collision threshold cause disproportionately higher danger.
- **Safety Threshold: 2.0s** - Based on widely accepted driver reaction time benchmarks used in traffic engineering and automotive safety standards (e.g., **AASHTO guidelines** recommend 2.5s for perception-reaction; 2.0s represents a commonly referenced optimistic baseline).

Comparative Risk Scores:

- **Standard HMI (TTC 1.5s):** $e^{3 \cdot (2.0-1.5)} = e^{1.5} \approx 4.48$
- **Accessibility HMI (TTC 2.0s):** $e^{3 \cdot (2.0-2.0)} = e^{0} = 1.0$

1. Distance vs Time

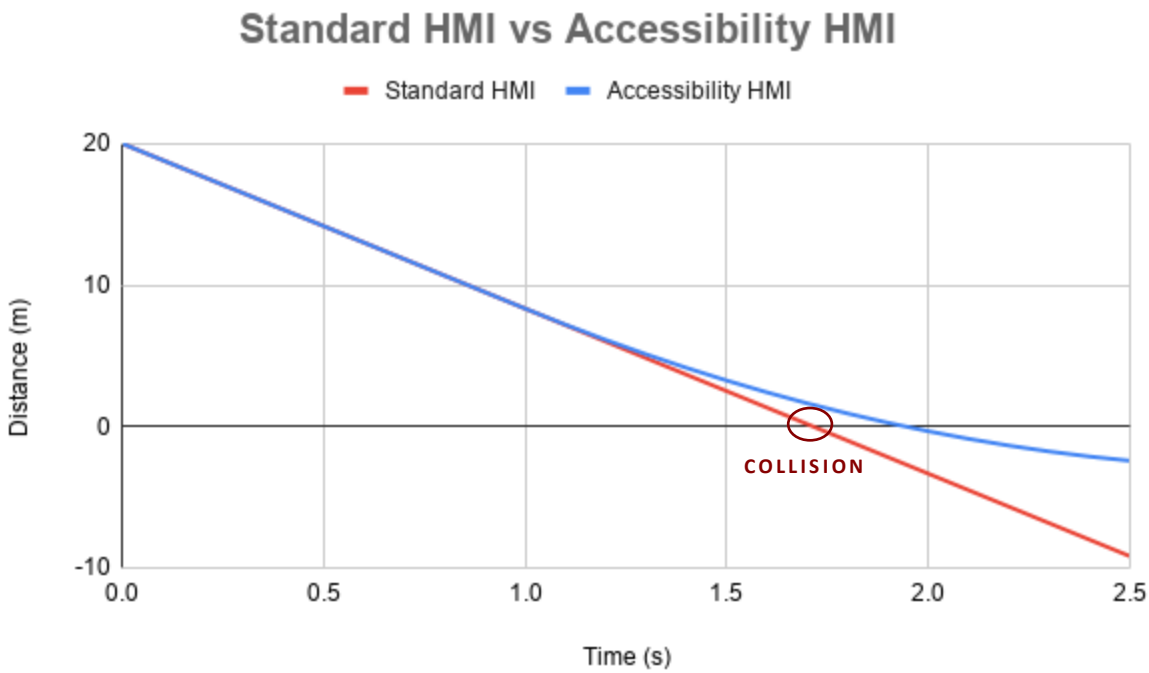

Figure 1: *Comparison of kinematics.* ***The Red line (Standard HMI)*** *reaches zero distance at* ***~1.7s****, resulting in a high-velocity collision.* ***The Blue line (Accessibility HMI)*** *demonstrates* the impact being **delayed to 1.94s** with significant deceleration.

2. Exponential Risk Model

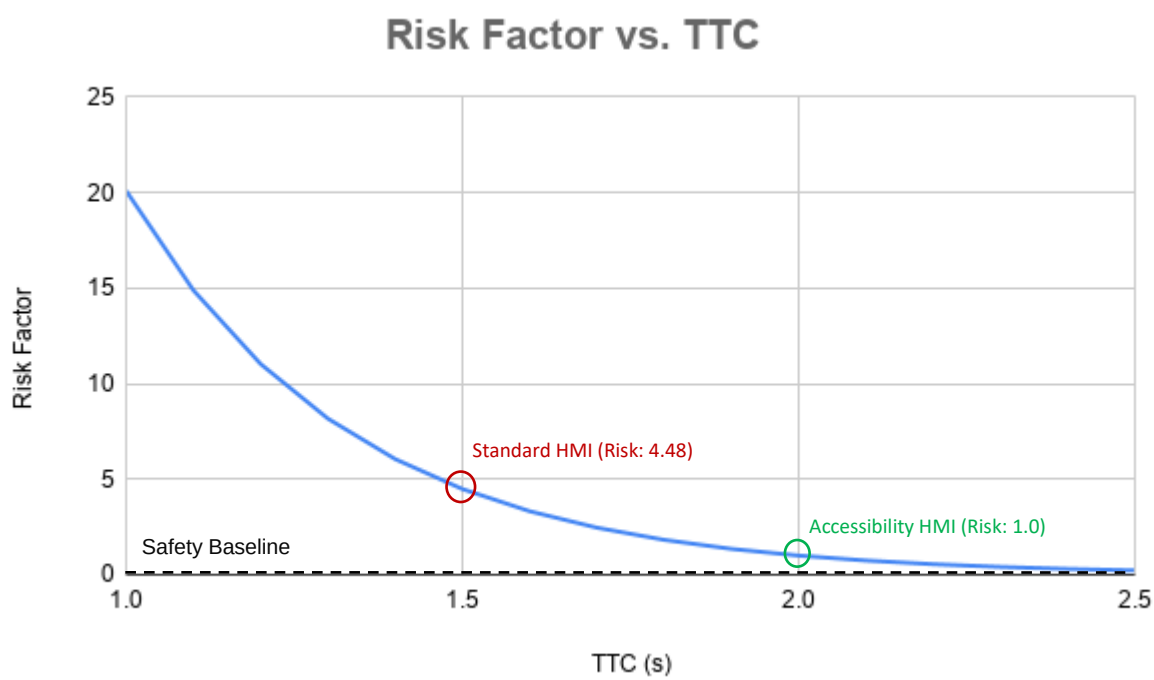

Figure 2: *Exponential collision risk model. The analysis shows that an increase in* ***effective TTC from 1.5s to 2.0s*** *moves the driver from an* ***exponential danger zone*** *(Risk Factor ≈ 4.48) to the* ***safety baseline*** *(Risk Factor = 1.0). Even minor improvements in perception time yield non-linear safety benefits.*

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5. Vision 2026: Extended Perception via V2X

The next evolution of this HMI is integration with **V2X (Vehicle-to-Everything)** infrastructure.

- **Scenario:** A smart camera on a roadside pole detects a hazard 300 meters ahead, beyond the driver's line of sight.
- **HMI Action:** Utilizing my existing framework, the V2X signal would trigger a **'Pre-Alert'** phase. While my current PoC focuses on the high-intensity Red 'DANGER' State for immediate hazards, the system logic allows for a lower-intensity state (e.g., yellow ambient lighting) to nudge the driver's attention long before the hazard is visible.
- **Impact:** This moves the reaction point hundreds of meters back, effectively reducing the risk factor to **zero** by enabling **preventive deceleration** instead of emergency braking.

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6. Study Limitations

While this analysis provides a robust framework for HMI safety, the following limitations should be noted:

- **Constant Deceleration:** This model assumes a constant deceleration rate of $\mathbf{6.0\ m/s^2}$. In real-world conditions, braking force varies based on **road surface (friction)**, **tire condition** and **ABS intervention**.
- **Prototype Environment:** The HMI PoC is a **browser-based simulation** designed for logic and UX research; it has not been tested on **ISO 26262-compliant** automotive hardware.
- **Reaction Time Assumptions:** The improvement in perception-reaction time (2.0s → 1.0s) is based on established academic research regarding **multimodal stimuli** and may vary across different demographic groups.

- **Hazard Behavior:** The model assumes the hazard (deer) maintains a constant velocity and linear path.

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7. Conclusion

The data confirms that **HMI design directly influences the physics of a crash**. By utilizing **Accessibility principles**, we treat the driver's temporary distraction as a **situational disability** and provide the necessary sensory tools to overcome it. My prototype proves that **multimodal interaction is not a luxury—it is a life-saving necessity** in modern automotive engineering.